

Trust-Aware Bio-Inspired Swarm Defense Using Physics-Based Simulation

1 Objectives

The main objective of this research work is to develop a fault-resilient multi-agent navigation framework which integrates bio-inspired immunological principles with deep reinforcement learning techniques. The specific objectives of this research are:

Resilient Swarm Navigation: To investigate and analyze whether a decentralized robot/drone swarm can successfully navigate towards a designated target location using learned multi-agent control policies (MAPPO) in a continuous environment with static obstacles.

Bio-Inspired Fault Detection: To design and develop a distributed “T-Cell” based trust mechanism which enables agents to autonomously identify and detect traitor drones by analyzing the inconsistencies between physical sensor data (LiDAR) and the communicated information received from neighboring agents.

Adversarial Robustness Evaluation: To perform systematic evaluation of the performance and resilience of the proposed framework under different adversarial intensities, specifically when 5%, 10%, 20%, and 30% of the total swarm consists of traitor agents.

Mission Success Assurance: To examine and verify whether the swarm can successfully maintain its cohesion and reach the target destination even when traitor drones actively attempt to deceive, disrupt, or collide with honest agents.

2 Proposed Methodology

In this study, we have proposed a Trust-Attentive Multi-Agent Proximal Policy Optimization (TA-MAPPO) framework. The proposed architecture combines a bio-inspired immune system approach for fault detection with a Centralized Training, Decentralized

Execution (CTDE) learning paradigm to ensure robust navigation even under adversarial conditions.

2.1 System Architecture and Simulation Environment

The system has been modeled as a Partially Observable Markov Decision Process (POMDP) within a continuous 2D physics simulation which is implemented using the PettingZoo library.

Agents: The swarm is comprised of N agents, which are divided into two categories — Honest Drones ($\mathcal{H}$) and Traitor Drones ($\mathcal{T}$).

Physics: All agents operate with continuous kinematics and are governed by velocity constraints and inertia.

Sensing Model: Each agent i in the swarm possesses two distinct sensing modalities:

1. **LiDAR (Trusted Physical Sensor):** This sensor detects the relative position p_{ij} and velocity v_{ij} of neighboring agents and static obstacles present in the environment. This data is physically grounded and therefore cannot be falsified by adversaries.
2. **Communication Channel (Untrusted Digital Channel):** Through this channel, agents receive self-reported states $\hat{p}_{ij}$ and $\hat{v}_{ij}$ which are broadcast by neighboring agents. However, traitor agents may falsify this communicated data in order to deceive the swarm.

2.2 Bio-Inspired T-Cell Based Fault Detection

Taking inspiration from T-Cell receptors present in the biological immune system, which bind to specific antigens to identify pathogens, each honest agent in our system runs a local, decentralized fault detection algorithm. This mechanism computes a Trust Score $T_{ij} \in [0, 1]$ for every neighboring agent j based on five behavioral “antigens” which serve as fault indicators.

2.2.1 Behavioral Fault Indicators (I_k)

At each time step t , the agent evaluates five binary flags as given below:

I_{dist} (**Deceptive Communication**): This flag is triggered if the discrepancy between the sensed distance obtained from LiDAR and the reported distance received through communication exceeds a predefined noise threshold ϵ .

$$I_{dist} = \mathbb{I}(|d_{ij}^{LiDAR} - d_{ij}^{Comm}| > \epsilon) \quad (1)$$

I_{col} (**Collision Intent**): This flag is triggered when a neighboring agent moves directly towards the current agent while being within critical proximity range.

I_{vel} (**Velocity Anomaly**): This flag is triggered if a neighbor’s velocity vector shows significant deviation from the local swarm average, indicating a cohesion violation.

I_{prox} (**Proximity Violation**): This flag is triggered when a neighboring agent violates the safety separation zones (comes too close) or exceeds the fragmentation limits (moves too far).

I_{dir} (**Directional Divergence**): This flag is triggered if a neighboring agent consistently moves in a direction which is opposite to the mission goal, indicating possible sabotage behavior.

2.2.2 Recursive Trust Update Mechanism

The Trust Score evolves over time using a recursive low-pass filter approach to prevent volatility caused by sensor noise:

$$T_{ij}(t) = \alpha T_{ij}(t-1) + (1-\alpha) \cdot \left(1 - \tanh\left(\sum I_k\right)\right) \quad (2)$$

where α is a memory factor (typically set to 0.95). When there is a high density of fault indicators, the value of T_{ij} decays toward 0, indicating low trust.

2.3 Trust-Attentive MAPPO Controller

The core navigation policy is learned using the MAPPO algorithm, which has been modified with a novel Feature Gating Mechanism to suppress adversarial data received from traitor agents.

2.3.1 Feature Gating Network (The “Trust Filter”)

To prevent the neural network from processing deceptive information coming from traitor agents, we have employed an attention-based gating architecture as described below:

Encoder: The raw neighbor observations o_j are first processed by a non-linear encoder (consisting of a Linear layer followed by ReLU activation) to extract a high-dimensional feature vector h_j .

Gating Mechanism: The extracted feature vector is then modulated by the Trust Score as follows:

$$\tilde{h}_j = h_j \odot T_{ij} \quad (3)$$

Effect: If $T_{ij} \approx 1$ (indicating an Honest agent), the features pass through without suppression. However, if $T_{ij} \approx 0$ (indicating a Traitor agent), the features are suppressed (essentially zeroed out). This mechanism ensures that the policy network effectively

“ignores” the traitor’s cooperative signals while still retaining physical awareness through LiDAR data for collision avoidance purposes.

2.3.2 CTDE Training Paradigm

Centralized Critic (V_ϕ): During the training phase, the Critic network has access to the global state information and ground-truth labels which identify which agents are traitors. It learns a Value Function that penalizes risky interactions with traitor agents.

Decentralized Actor (π_θ): During the execution phase, the Actor network operates solely on local, trust-gated observations to generate velocity commands (v_x, v_y) for each agent.

2.4 Reward Function Design

The reinforcement learning objective $J(\theta)$ aims to maximize the cumulative return which is defined by:

$$R_t = R_{goal} + R_{safe} + R_{group} \quad (4)$$

R_{goal} : This component provides a positive reward for reducing the distance to the target location and gives a large bonus when the agent successfully arrives at the destination.

R_{safe} : This component imposes a heavy penalty (set to -100) for any collision events with obstacles or other drones in the environment.

R_{group} : This is a cohesion reward which encourages agents to maintain formation with only trusted neighbors, thereby incentivizing the isolation of untrusted agents from the swarm.

2.5 Adversarial Traitor Modeling

To evaluate the resilience of our proposed system, a subset of agents (ranging from 5% to 30% of the total swarm) are designated as Traitors. These traitor agents follow a fixed adversarial policy with the following behaviors:

Deception: They broadcast false coordinate information to other agents, where the reported position is given by:

$$\hat{p} = p_{true} + \text{Noise} \quad (5)$$

Aggression: They actively accelerate towards honest agents with the intention to induce collisions.

Disruption: They move efficiently in directions that are opposite to the swarm goal in order to fragment the group and disrupt cohesion.